

Control of a Radar Seeker using Fuzzy Type-2 PID Controller

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Abstract—A two-axis seeker must be stabilized from the missile body motions in different directions. Tracking the target using a missile is depends on the performance stability of the seeker. For this purpose, a dynamic model of a seeker and missile motion is needed. So in this paper first we implement a mathematical model of the seeker and then stabilize it with respect to the motion of the missile using Fuzzy Type-2 PID controller.

Keywords— Fuzzy Type2, Missile, PID, Seeker, Stabilization

I. INTRODUCTION

There are different types of seekers that are widely used. By analyzing the reflected signals, the seeker can determine the target's location, speed, and direction. Usually, a seeker is mounted in front of a missile, and the seeker axis is aligned with the missile body axis [1]. Radar seekers are effective for long-range targeting and can operate in various weather conditions. The radar seeker is the core component of missiles for intercepting and tracking radiation source [2]. Radar servo mechanism plays an important role in isolating the missile body motions in order to stabilize the seeker line of sight. The precision of servo performance and rejection of disturbance are necessary to stabilize the image of the target transmitted to control panel and maintain a target's line of sight (LOS) [3]. A seeker consists of two gimbals that can be moved in yaw and pitch directions by two motors. The angles related to the movement of the gimbals in yaw and pitch directions are measured by two encoders, and two rate gyros are placed in the system in order to stabilize the movement of the seeker. A radar seeker including its motors and sensors is shown in Figure 1 [4]. Such systems usually have the tracking control loop and the stabilization control loop as an inner loop. The main function of these control loops is to keep the boresight of system unchangeable and also stabilize the line of sight of the seeker. Especially, the stabilization control loop must provide excellent body motion decoupling performance for good target tracking of missiles [5].

In different papers, various methods have been used to control the stability of the seeker according to the type of the seeker. In [6] a sliding mode method is used for stabilization of a roll-pitch seeker. In [7] a radar seeker is stabilized with a Backstepping sliding mode. Fuzzy PID control is used in [8] to stabilize a two-axis gimbal seeker. Due to the nature of the seeker and the conditions of its use, most of the methods cannot fully consider the system uncertainty in the control loop, so in this paper we use Fuzzy PID Type-2 to control the system by considering the uncertainties in the system.

II. DYNAMIC MODEL OF A TWO-AXIS SEEKER

A dynamic model for a radar seeker is developed as follows [9]:

The coordinate systems of the seeker and missile body can be shown like Figure 2.

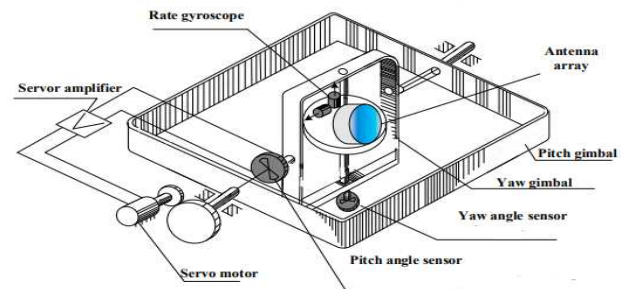


Figure 1. Two axis radar seeker

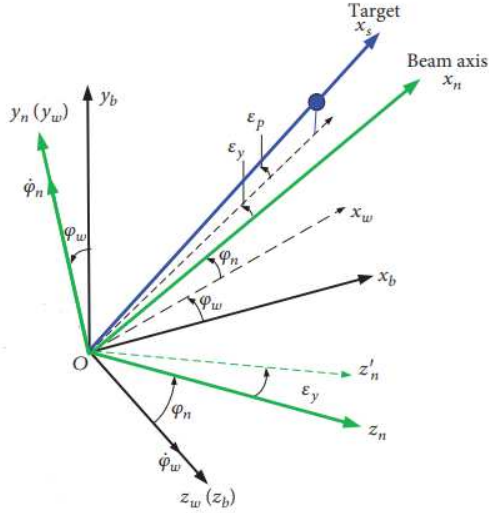


Figure 2. Coordinate system

Coordinate systems and transformation relations are introduced as follows:

$Ox_i y_i z_i$: Inertial coordinate system: Ox_i congruous to the flat earth surface, Oy_i is in perpendicular direction, the direction of Oz_i can be obtained according to the other two axes.

$Ox_b y_b z_b$: Coordinate system of Missile body: Ox_b congruous to the longitudinal axis of the missile and points to the front of the missile. Oy_b is perpendicular to the Ox_b and its direction is upwards.

$Ox_w y_w z_w$: Outer gimbal coordinate system: Ox_w direction points towards the target, Oz_w congruous to Oz_b of the above coordinate system (missile body), and Oy_w can be obtained according to the other two axes.

$Ox_n y_n z_n$: Coordinate system of the Inner gimbal: Ox_n direction points towards the target and congruous to the beam direction, Oy_n congruous to Oy_w of outer gimbal coordinate system, and Oz_n can be obtained according to the other two axes.

$Ox_s y_s z_s$: Line Of Sight coordinate system (LOS): Ox_s direction points towards the target, Oy_s is perpendicular to Ox_s and pointing up its direction is upwards, and Oz_s can be obtained according to the other two axes.

From $Ox_b y_b z_b$ to $Ox_w y_w z_w$, Oz_b is axis of rotation; ϕ_w is the angle of rotation. From $Ox_w y_w z_w$ to $Ox_n y_n z_n$, Oy_w is the axis and the parameter ϕ_n is the angle of rotation. From $Ox_n y_n z_n$ to $Ox_s y_s z_s$, first, the axis that corresponds to the rotation is Oy_n and the angle that corresponds to the rotation is ε_y , and then the axis that corresponds to the rotation is Oz'_n and the angle that corresponds to the rotation is ε_p .

ψ , θ and γ are missile body yaw, pitch and roll angles. q_y and q_p are the angle parameters of Line Of Sight in the yaw and pitch directions. ϕ_w and ϕ_n are the angles parameters of the gimbal in the pitch and yaw directions. ε_y and ε_p are the parameters of the seeker that corresponds to the off-centered angles of antenna beam in the yaw and pitch directions.

The transformation matrices of different transformation can be written as follows:

$$L_{b \rightarrow \omega}(\phi_w) = \begin{bmatrix} \cos \phi_w & \sin \phi_w & 0 \\ -\sin \phi_w & \cos \phi_w & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

$$L^{-1}_{i \rightarrow s} = \begin{bmatrix} \cos q_y \cos q_p & -\cos q_y \sin q_p & \sin q_y \\ \sin q_p & \cos q_p & 0 \\ -\sin q_y \cos q_p & \sin q_y \sin q_p & \cos q_y \end{bmatrix} \quad (2)$$

$$L_{i \rightarrow b} = \begin{bmatrix} \cos \theta \cos \psi & \sin \theta \\ -\sin \theta \cos \psi \cos \gamma + \sin \psi \sin \gamma & \cos \theta \cos \gamma \\ \sin \theta \cos \psi \sin \gamma + \sin \psi \cos \gamma & -\cos \theta \sin \gamma \\ -\cos \theta \sin \psi & \sin \theta \sin \psi \cos \gamma + \cos \psi \sin \gamma \\ \sin \theta \sin \psi \sin \gamma + \cos \psi \cos \gamma & \end{bmatrix} \quad (3)$$

$$L_{b \rightarrow n} = \begin{bmatrix} \cos \phi_n \cos \phi_w & \sin \phi_w & -\sin \phi_n \cos \phi_w \\ -\cos \phi_n \sin \phi_w & \cos \phi_w & \sin \phi_n \sin \phi_w \\ \sin \phi_n & 0 & \cos \phi_n \end{bmatrix} \quad (4)$$

If the antenna boresight unit vector in LOS coordinate system ($\vec{a}$) is taken as $[1 \ 0 \ 0]^T$ and it is along the line of sight,

we have:

$$\begin{aligned} \varepsilon_y &= \arctan\left(-\frac{z_{n2}}{x_{n2}}\right) \\ \varepsilon_p &= \arctan\left(\frac{y_{n2}}{x_{n2}}\right) \end{aligned} \quad (5)$$

Where:

$$[x_{n2} \ y_{n2} \ z_{n2}]^T = L_{b \rightarrow n} L_{i \rightarrow b} L^{-1}_{i \rightarrow s} \vec{a} \quad (6)$$

According to figure 2, we have:

$$\omega_b^{b/i} = [\omega_{bx} \ \omega_{by} \ \omega_{bz}]^T \quad (7)$$

Where:

b/i denotes missile body with respect to the inertial space.

$$\omega_n^{n/b} = [-\dot{\phi}_w \sin \phi_n \ \dot{\phi}_n \ \dot{\phi}_w \cos \phi_n]^T \quad (8)$$

$$\begin{aligned} \omega_n^{b/i} &= L_{\omega \rightarrow n}(\phi_n) L_{b \rightarrow \omega}(\phi_w) \omega_b^{b/i} \\ &= \begin{bmatrix} (\omega_{bx} \cos \phi_w + \omega_{by} \sin \phi_w) \cos \phi_n - \omega_{bz} \sin \phi_n \\ -\omega_{bx} \sin \phi_w + \omega_{by} \cos \phi_w \\ (\omega_{bx} \cos \phi_w + \omega_{by} \sin \phi_w) \sin \phi_n + \omega_{bz} \cos \phi_n \end{bmatrix} \end{aligned} \quad (9)$$

Using (7), (8) and (9):

$$\omega_n^{n/i} = \omega_n^{n/b} + \omega_n^{b/i} = [\omega_{nx}^{n/i} \ \omega_{ny}^{n/i} \ \omega_{nz}^{n/i}]^T \quad (10)$$

Where:

$$\begin{aligned}
\omega_{nx}^{n/i} &= (\omega_{bx}\cos\varphi_\omega + \omega_{by}\sin\varphi_\omega)\cos\varphi_n - (\omega_{bz} + \dot{\varphi}_\omega)\sin\varphi_n \\
\omega_{ny}^{n/i} &= -\omega_{bx}\sin\varphi_\omega + \omega_{by}\cos\varphi_\omega + \dot{\varphi}_n \\
\omega_{nz}^{n/i} &= (\omega_{bx}\cos\varphi_\omega + \omega_{by}\sin\varphi_\omega)\sin\varphi_n + (\omega_{bz} + \dot{\varphi}_\omega)\cos\varphi_n
\end{aligned} \quad (11)$$

Similar to the previous method, regarding the external gimbal, it can be written as follows:

$$\omega_\omega^{b/i} = \begin{bmatrix} \omega_{bx}\cos\varphi_\omega + \omega_{by}\sin\varphi_\omega \\ -\omega_{bx}\sin\varphi_\omega + \omega_{by}\cos\varphi_\omega \\ \omega_{bz} \end{bmatrix} \quad (12)$$

$$\begin{aligned}
\omega_\omega^{\omega/i} &= \begin{bmatrix} \omega_{\omega x}^{\omega/i} \\ \omega_{\omega y}^{\omega/i} \\ \omega_{\omega z}^{\omega/i} \end{bmatrix} = \omega_\omega^{\omega/b} + \omega_\omega^{b/i} \\
&= \begin{bmatrix} \omega_{bx}\cos\varphi_\omega + \omega_{by}\sin\varphi_\omega \\ -\omega_{bx}\sin\varphi_\omega + \omega_{by}\cos\varphi_\omega \\ \omega_{bz} + \dot{\varphi}_\omega \end{bmatrix}
\end{aligned} \quad (13)$$

So

$$\begin{aligned}
\dot{\omega}_{nx}^{n/i} &= (\dot{\omega}_{\omega x}^{\omega/i} - \omega_{\omega z}^{\omega/i} \dot{\varphi}_n)\cos\varphi_n - \omega_{\omega x}^{\omega/i} \dot{\varphi}_n\sin\varphi_n - \dot{\omega}_{\omega z}^{\omega/i}\sin\varphi_n \\
\dot{\omega}_{nz}^{n/i} &= (\dot{\omega}_{\omega x}^{\omega/i} - \omega_{\omega z}^{\omega/i} \dot{\varphi}_n)\sin\varphi_n + \omega_{\omega x}^{\omega/i} \dot{\varphi}_n\cos\varphi_n + \dot{\omega}_{\omega z}^{\omega/i}\cos\varphi_n
\end{aligned} \quad (14)$$

To improve the tracking performance, it is necessary to consider the disturbance torques acting on the gimbal axis [10].

To investigate the effect of different torques on the system, Newton's laws are applied to internal and external gimbals.

$$\begin{aligned}
\sum \vec{M}_n &= [M_{nx} \ M_{ny} \ M_{nz}]^T \\
&= \frac{d\vec{H}_n}{dt} + \vec{\omega}_n^{n/i} \times \vec{H}_n + \vec{\rho}_n \times \vec{a}_n m_n \\
\vec{H}_n &= \begin{bmatrix} H_{nx} \\ H_{ny} \\ H_{nz} \end{bmatrix} = \begin{bmatrix} J_{nxx} & 0 & 0 \\ 0 & J_{nyy} & 0 \\ 0 & 0 & J_{nzz} \end{bmatrix} \begin{bmatrix} \omega_{nx}^{n/i} \\ \omega_{ny}^{n/i} \\ \omega_{nz}^{n/i} \end{bmatrix}
\end{aligned} \quad (15)$$

Where $\vec{H}_n$ is the angular momentum, J_n is the inertia term, $\vec{\rho}_n \times \vec{a}_n m_n$ is the cross product of the moment arm and the mass of the inner gimbal times the specific force acceleration.

Because the inner gimbal only has freedom in yaw direction, M_{ny} term is taken.

$$M_{ny} = M_{ny}^{CT} - K_{Sn}\varphi_n - K_{Vn}\dot{\varphi}_n - K_{couln}\left(\frac{\dot{\varphi}_n}{|\dot{\varphi}_n|}\right) \quad (16)$$

Where M_{ny}^{CT} is the inner gimbal controller torque and K_{Sn} , K_{Vn} and K_{couln} represent the factors of the spring, viscous, and Coulomb friction torque, respectively.

And for the outer gimbal and according to the rotational freedom in pitch direction:

$$M_{\omega z} = M_{\omega z}^{CT} - K_{S\omega}\varphi_\omega - K_{V\omega}\dot{\varphi}_\omega - K_{coul\omega}\left(\frac{\dot{\varphi}_\omega}{|\dot{\varphi}_\omega|}\right) - (-M_{nx}\sin\varphi_n + M_{nz}\cos\varphi_n) \quad (17)$$

In above equation, the last term represents the torque that is generated due to the effect of the inner gimbal on the outer gimbal.

According to these equations:

$$\begin{aligned}
\dot{\omega}_{ny}^{n/i} &= \frac{J_{11}F_1 + J_{12}F_2 + J_{11}M_{ny} + J_{12}M_{cc}}{J_{eq}} \\
\dot{\omega}_{\omega z}^{\omega/i} &= \frac{J_{12}F_1 + J_{nyy}F_2 + J_{12}M_{ny} + J_{nyy}M_{cc}}{J_{eq}}
\end{aligned} \quad (18)$$

If the inertia product and the mass imbalance moment are zero, then:

$$\begin{aligned}
J_{11} &= J_{\omega zz} + J_{nxx}\sin^2\varphi_n + J_{nzz}\cos^2\varphi_n \\
J_{12} &= 0 \\
F_1 &= -\omega_{nz}^{n/i}H_{nx} + \omega_{nx}^{n/i}H_{nz} \\
F_2 &= (J_{nxx}(\dot{\omega}_{nx}^{n/i} + \dot{\omega}_{\omega z}^{\omega/i}\sin\varphi_n) + \omega_{ny}^{n/i}H_{nz} - \omega_{nz}^{n/i}H_{ny})\sin\varphi_n \\
&\quad - (J_{nzz}(\dot{\omega}_{nz}^{n/i} + \dot{\omega}_{\omega z}^{\omega/i}\cos\varphi_n) + \omega_{nx}^{n/i}H_{ny} - \omega_{ny}^{n/i}H_{nx})\cos\varphi_n \\
&\quad - \omega_{\omega x}^{\omega/i}H_{\omega y} + \omega_{\omega y}^{\omega/i}H_{\omega x} \\
J_{eq} &= J_{nyy}J_{\omega zz} + J_{nxx}\sin^2\varphi_n + J_{nzz}\cos^2\varphi_n
\end{aligned} \quad (19)$$

III. SEEKER CONTROL DESIGN

To control the seeker performance, the control system is composed of two control loops. In order to control and navigate seeker systems in elevation and azimuth axes, two controlling loops, stabilizer and tracker, have been used [11]. In this paper we use these two loops as Figure 3.

We use a Fuzzy Type-2 PID controller for both stability and tracking control loop to stabilize the performance of the seeker and keep the antenna off-centered angle unchangeable.

Due to the existence of uncertainty in these systems, we use fuzzy logic to handle these uncertainties. Basically, fuzzy controller comprises of four main components; fuzzification interface, knowledge base, inference mechanism and defuzzification interface [12].

One of the main characteristics of the fuzzy logic is that it handles the uncertainties of the system. In systems with uncertainties, the use of fuzzy logic in system analysis and control, seems to be more logical than crisp systems, while if these uncertainties are high, the use of crisp membership functions (MF) cannot cover all uncertainties, so the fuzzy type-2 algorithm is used. [13]. Type-2 FLCs that use type-2 fuzzy sets can handle such uncertainties to produce a better performance [14].

As shown in Figure 3, the angles of missile body enter the antenna off-centered angle block and after the calculation of antenna off-centered angles, these angles can be amplified by the gain to become LOS rate angle.

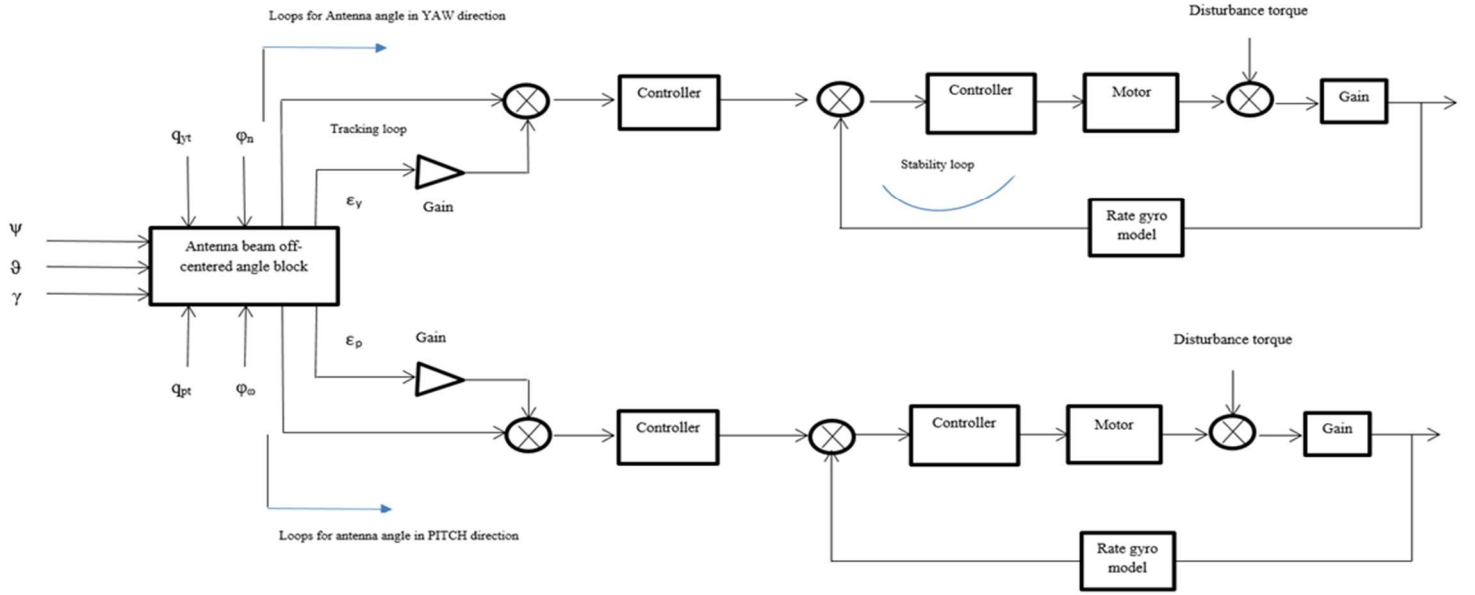


Figure 3. Tracking and Stability control loop

For this loop called tracking loop, we use a fuzzy type2 controller keeping the off-centered angles stable.

We use the model of torque motor [9] in stability loop in order to investigate the influence of the torque that is applied to the system as a result of disturbance; the disturbance torque is added to the model. In stability loop, to sample the output, the model of rate gyro is used in feedback.

The fuzzy Type-2 PID controller has two inputs and three outputs. The inputs are error (e_r) and error deviation (der) and the outputs are the variation of controller parameters (ΔK_p , ΔK_i , ΔK_d) [15].

A. Fuzzification

We can choose seven subsets for the variables of the system, these subsets are: {VN, LN, NO, ZO, PO, LP, VP}. These subsets represent Very Negative, Less Negative, Negative, Zero, Positive, Less Positive, and Very Positive. The domain of discourse for two inputs and three outputs variables is $\{-3, -2, -1, 0, 1, 2, 3\}$.

B. Fuzzy Inference Rule

We used the Gaussian type of membership function as shown in Figure 4. The fuzzy type-2 rule base tables of three parameters of PID controller were obtained as table I to table III.

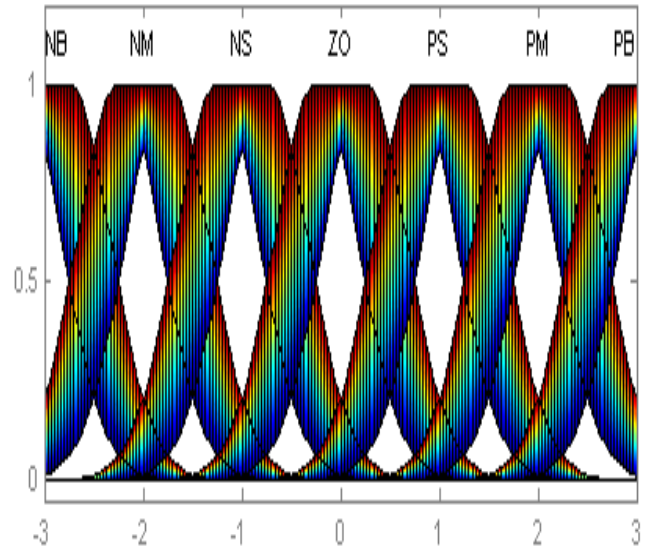


Figure 4. Gaussian membership function for Inputs and Outputs

TABLE I. ΔK_P RULE BASE

$\frac{\text{der}}{\Delta K_P}$	VN	LN	NO	ZO	PO	LP	VP
VN	VP	VP	LP	LP	PO	ZO	ZO
LN	VP	VP	LP	PO	PO	ZO	NO
NO	LP	LP	LP	PO	ZO	NO	NO
ZO	LP	LP	PO	ZO	NO	LN	LN
PO	PO	PO	ZO	NO	NO	LN	LN
LP	PO	ZO	NO	LN	LN	LN	VN
VP	ZO	ZO	LN	LN	LN	VN	VN

TABLE II. ΔK_I RULE BASE

$\frac{\text{der}}{\Delta K_I}$	VN	LN	NO	ZO	PO	LP	VP
VN	VN	VN	LN	LN	NO	ZO	ZO
LN	VN	VN	LN	NO	NO	ZO	ZO
NO	VN	LN	NO	NO	ZO	PO	PO
ZO	LN	LN	NO	ZO	PO	LP	LP
PO	LN	NO	ZO	PO	PO	LP	VP
LP	ZO	ZO	PO	PO	LP	VP	VP
VP	ZO	ZO	PO	LP	LP	VP	VP

TABLE III. ΔK_D RULE BASE

$\frac{\text{der}}{\Delta K_D}$	VN	LN	NO	ZO	PO	LP	VP
VN	VN	VN	LN	LN	NO	ZO	ZO
LN	VN	VN	LN	LN	NO	ZO	ZO
NO	LN	VN	NO	NO	ZO	LP	PO
ZO	LN	LN	NO	ZO	PO	LP	LP
PO	LN	NO	ZO	PO	PO	LP	LP
LP	ZO	ZO	LP	PO	LP	VP	VP
VP	ZO	ZO	PO	LP	LP	VP	VP

For the fuzzy Inference system, we chose Mamdani algorithms. The output of the fuzzy2 controller is the change in the control signal, so the fuzzy type-2 controller parameters were calculated as follow:
 $K_P + \Delta K_P$, $K_I + \Delta K_I$ and $K_D + \Delta K_D$

IV. SIMULATION RESULTS

First, we enter a sine wave as the yaw angle and the pitch angle of the missile separately while the input LOS angle is zero and check the performance of the system without a controller. In an ideal condition the off-centered angle should be zero and the gimbal motion should be against the missile motion to keep the LOS stable.

In the following figures, the diagrams of the motion of the missile, the antenna beam angle, the line of sight angle and the seeker movement angle in yaw and pitch directions are shown.

The output without controller is shown in figure 5 and figure 6.

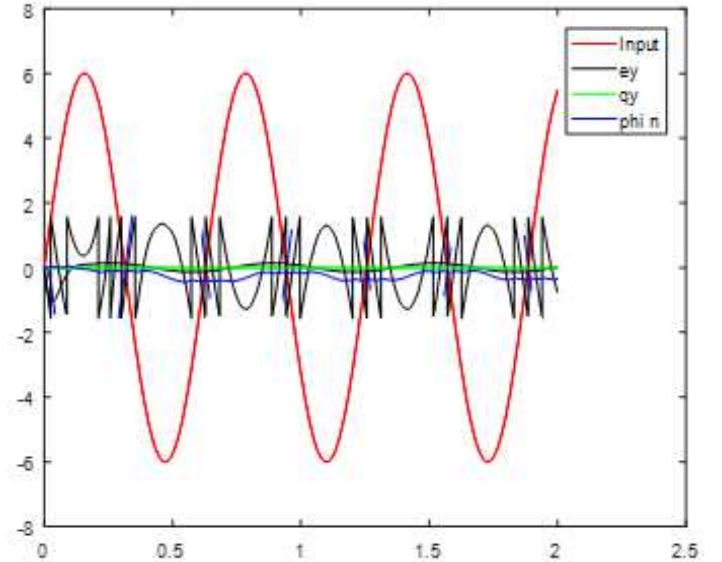


Figure 5. Missile Yaw input without controller

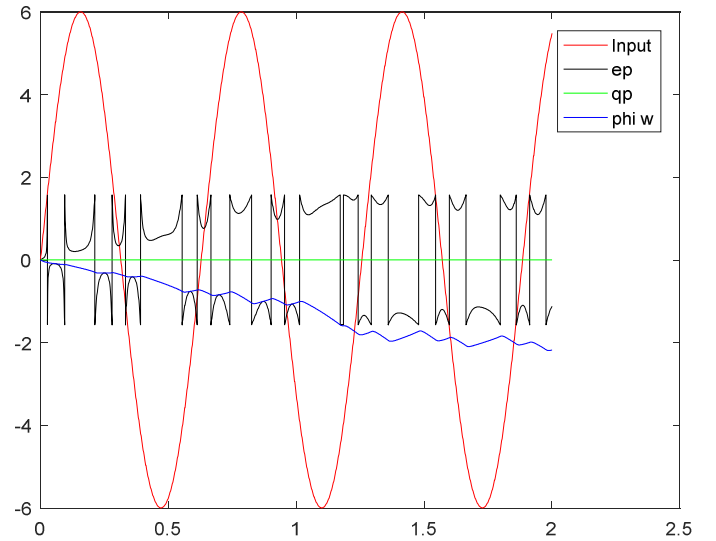


Figure 6. Missile Pitch input without controller

Figure 7 and figure 8 show the output using the fuzzy type2 PID controller in stability and tracking control loop.

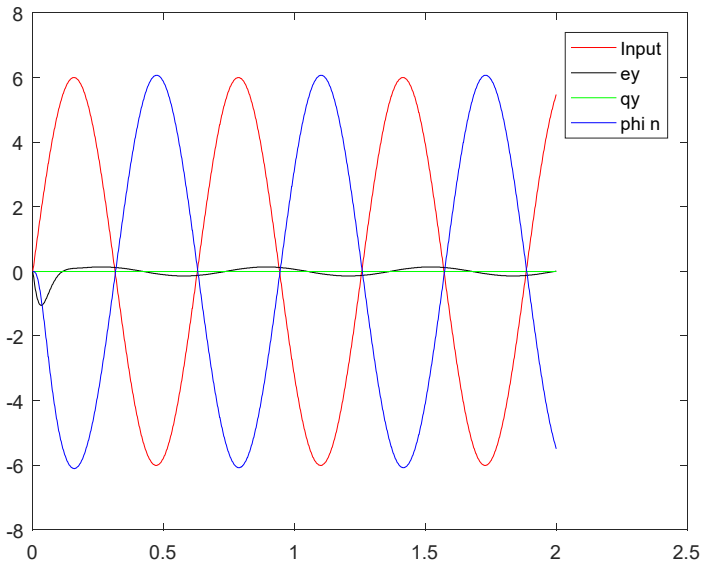


Figure 7. Missile Yaw input with controller

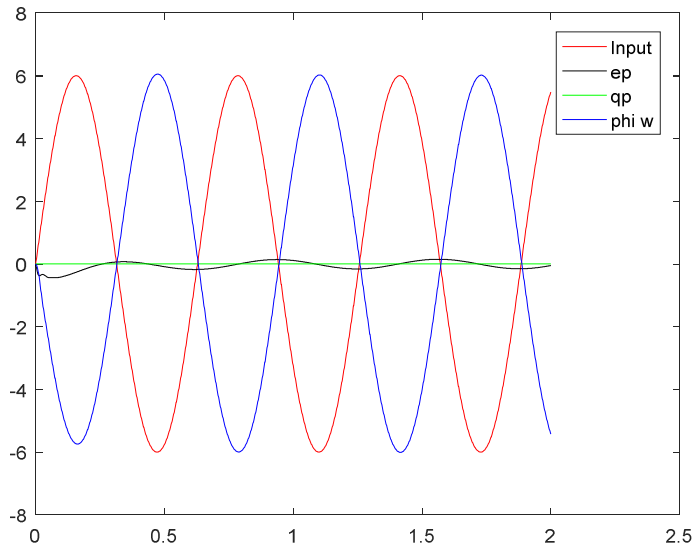


Figure 8. Missile Pitch input with controller

V. CONCLUSION

As shown in simulation results, when the input is given to the model as missile movement angles, the seeker movement must be in the opposite direction of the missile movement in order to keep the LOS angles constant. The results with the controller show the role of fuzzy PID controller in keeping antenna beam angles constant. The fuzzy controller makes the angles of the line of sight and the antenna beam remain almost zero and unchanged.

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